

Higher Category Theory

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1 Introduction

The traditional category theory is about objects and morphisms. In these notes, we want to generalize category into something that has more structures. Consider the following scenario: The category **Top** has objects that are topological spaces and morphisms that are continuous maps between two topological spaces. We have homotopy between two continuous maps with the same ends of topological spaces. More precisely, let X and Y be two topological spaces and $f, g : X \rightarrow Y$ be two continuous maps. Then, we can talk about a homotopy F between f and g , Which can be thought of as a 'morphism' from f to g . This is a morphism of morphisms. Apparently, it has the hierarchy one level higher than the morphism in the category **Top**. To distinguish these two different kinds of 'morphisms', we call the original morphism in a category 1-morphism and this 'morphism of morphisms' 2-morphism.

This enlightens us to generalize these objects and morphisms into abstract stuff, called cells. Seemingly, here the whole stuff has more things than a normal category. If we take the homotopy map into consideration, the whole entity has more 'morphisms' than the morphisms of a category. This entity deserves a name, called 2-category.

Definition 1.1: 2-category